

Adversarial Fairness

Qualitative & Analytic Report

Dataset: MobilitySurvey

2026-06-27

Accuracy: 85.7% -> 66.5%
Iterations: 1
sex: PASS — P-rule 92% $\geq$ 80%
origin_zone: PASS — P-rule 85% $\geq$ 80%
schooling: PASS — P-rule 86% $\geq$ 80%

Qualitative analysis (LLM)

The disadvantaged group for each sensitive attribute are females, individuals from non-coastal origin zones, and those with lower schooling levels. The model's predicted-positive rate gaps between these groups and their respective advantaged counterparts are 8.3 percentage points (56.0% vs 47.7%) for sex, 9.8 percentage points (57.8% vs 48.0%) for origin zone, and 8.3 percentage points (57.9% vs 49.6%) for schooling. The model mirrors the actual outcome-rate gaps between these groups, with females having a significantly higher actual outcome rate than males, individuals from coastal zones having a higher actual outcome rate than those from non-coastal zones, and those with higher schooling levels having a much higher actual outcome rate than those with lower schooling levels. Debiasing improved the P-rule for sex by 35.3 percentage points (from 57.0% to 92.3%), but at the cost of reducing accuracy from 85.7% to 66.5%. The gaps in predicted-positive rates are not designed, but rather flow through correlated proxies such as origin zone and schooling level, which may be related to unobserved factors that influence entry status.

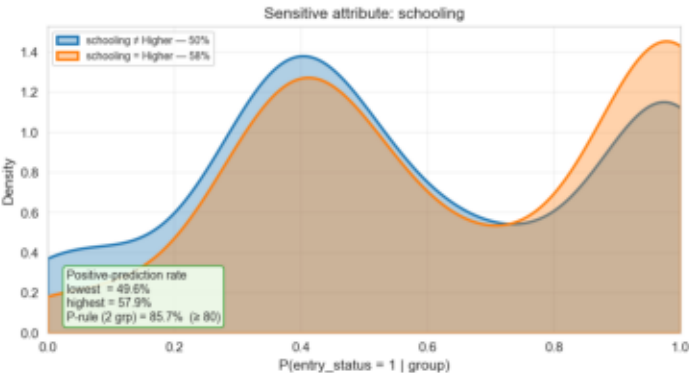
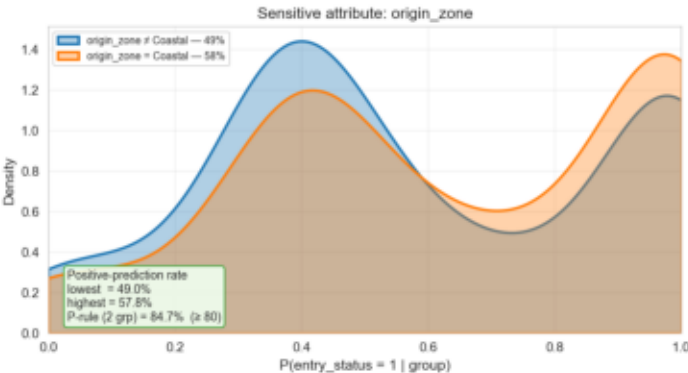
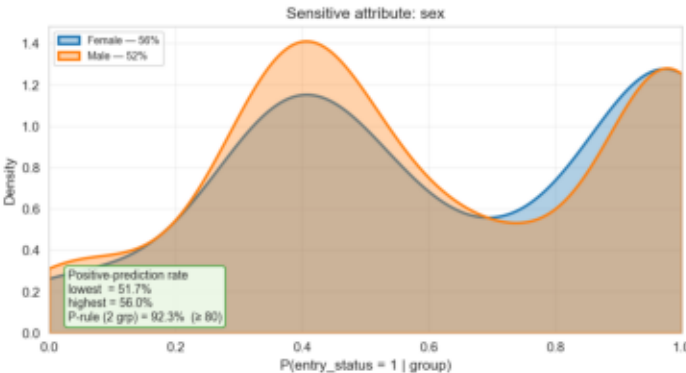
Fairness metrics (before -> after)

attribute	P-rule	DP	EoDD	EoOpp
sex	57.0 -> 92.3	36.1 -> 4.3	12.8 -> 7.8	13.3 -> 10.8
origin_zone	63.8 -> 84.8	25.9 -> 8.8	9.9 -> 14.7	7.6 -> 9.8
schooling	39.0 -> 85.7	55.7 -> 8.3	27.5 -> 8.4	22.1 -> 16.2

P-rule higher is fairer; DP / EoDD / EoOpp lower is fairer.

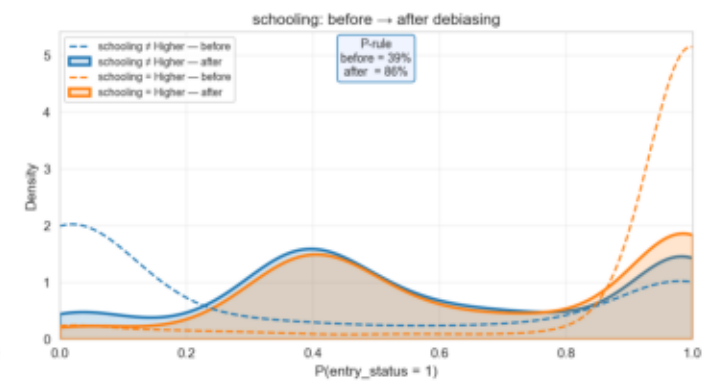
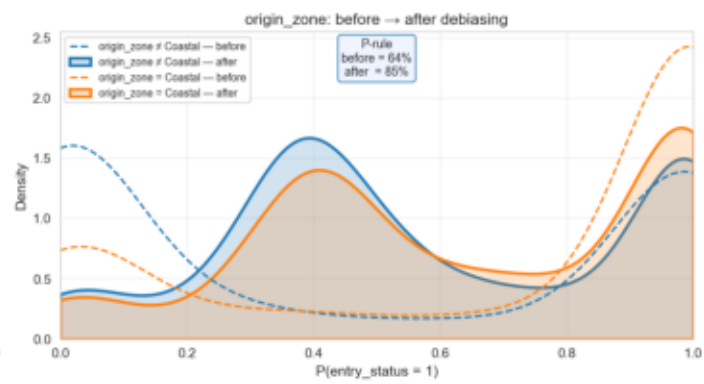
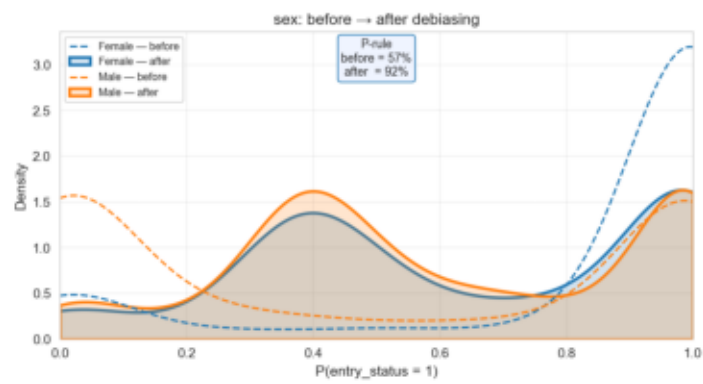
Predicted-outcome distribution by group

MobilitySurvey — predicted P(entry_status = 1) by sensitive group



Before → after debiasing

MobilitySurvey — before → after debiasing



Equalized odds — before vs after

MobilitySurvey — equalized odds before vs after

Equalized odds: before vs after debiasing

